Experiment No.: 10

Date of Performance :
Date of Submission :

Aim: Study of cellular network parameters using smartphones(signal strength,network type 2G/3G/4G/5G).

Required Equipment / Software :

- Android or iOS smartphone with an active SIM card and cellular network connection.
- Cellular signal/network information application(Netmonster etc..) or the phone's built-in network information screen.
- Mobile data enabled on the smartphone.
- Different measurement locations such as classroom, corridor, laboratory, ground floor, upper floor and open area.
- Notebook/observation sheet for recording network type and signal measurements.
- Computer/laptop with MATLAB for data analysis and plotting.

Theory :

A cellular network divides a geographical area into cells. Each cell is served by a base station or radio access network equipment. A smartphone communicates with the cellular network through radio signals. The quality of communication depends on parameters such as received signal strength, network technology, frequency band, cell identity and radio conditions.

Signal Strength

Cellular signal strength is commonly reported in dBm. The value is normally negative. A value closer to 0 dBm represents a stronger received signal. For example, -70 dBm is stronger than -100 dBm. The exact interpretation depends on the cellular technology and measurement reported by the device.

For 4G/LTE and 5G, devices may also provide RSRP (Reference Signal Received Power), RSRQ (Reference Signal Received Quality), and SINR. For 2G, RSSI is commonly used, while 3G may provide RSCP and Ec/No depending on the device.

Cellular Network Types

2G: Second-generation cellular technology, commonly associated with GSM and services such as GPRS/EDGE. It supports voice, SMS and relatively low-speed data.

3G: Third-generation cellular technology providing higher data capability than 2G and supporting mobile Internet and multimedia services.

4G/LTE: Fourth-generation technology based mainly on LTE, providing higher data rates and lower latency than earlier generations.

5G: Fifth-generation cellular technology designed to support high data rates, lower latency and high device density, depending on spectrum, coverage and network configuration.

Factors Affecting Cellular Signal Strength

Distance from the serving cell/base station.

Walls, floors, doors and other building materials.

Buildings, trees, vehicles and other obstructions.

Indoor versus outdoor location.

Frequency band and propagation characteristics.

Radio interference and network traffic.

Smartphone position, orientation and antenna characteristics.

Experimental Setup

The smartphone is used as the measurement device. At each selected location, the network information screen/application is opened and the current network technology and signal measurement are recorded. The smartphone should remain stationary for a short period before recording each reading.

Smartphone → Cellular Radio Network → Serving Cell / Base Station

Procedure :

1. Insert an active SIM card into the smartphone and ensure that cellular service is available.
2. Enable mobile data and, if required, select a suitable preferred network mode such as 2G/3G/4G/5G auto.

3. Open the phone's network information screen or a suitable cellular measurement application.

4. Select different measurement locations such as classroom, corridor, laboratory, ground floor, upper floor and outdoor area.

5. At each location, keep the phone stationary for a short time before recording the reading.

6. Record network type and the signal measurement in dBm or the technology-specific metric shown by the phone.

7. If available, record RSRP, RSRQ, SINR, cell ID and frequency band.

8. Repeat measurements two or three times where possible and calculate the

average. 9. Enter the collected data into MATLAB.

10. Plot signal strength against measurement location and compare the observed cellular technologies.

Observation Table :

S.No	Location	Network Type	Signal Strength (dBm)	RSRP (dBm)*	RSRQ (dB)*	SINR (dB)*
1	Ground Floor	4G/LTE	-82	-95	-12	8
2	Corridor	4G/LTE	-76	-89	-10	12
3	Near Window	5G	-69	-82	-8	18
4	Open Area	5G	-61	-75	-6	25
5	Basement	3G	-96	-105	-15	3
6	Weak Coverage Area	2G	-101	-110	-18	0
7	Upper Floor	4G/LTE	-73	-86	-7	15

Calculation :
if signal-strength readings at three locations are -82 dBm, -76 dBm and -69 dBm:

Average signal strength = [(-82) + (-76) + (-69)] / 3 = -75.67 dBm

Because the values are negative, a less-negative value indicates a stronger received signal. Thus, -69 dBm represents a stronger signal than -82 dBm.

Program :

% Study of Cellular Network Parameters Using a Smartphone

```
% Signal strength versus measurement location
clc;
clear;
close all;
% Sample measurements obtained from a smartphone
location = {'Classroom','Corridor','Near Window','Open Area', ...
 'Basement','Lift/Weak Area','Upper Floor','Entrance'};

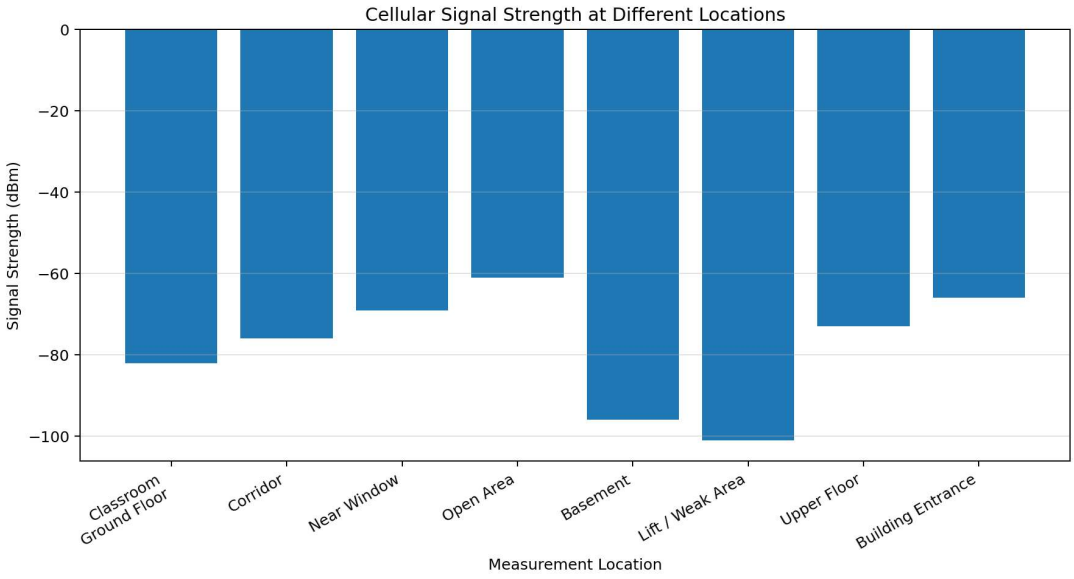
networkType = {'4G/LTE','4G/LTE','5G','5G','3G','2G','4G/LTE','5G'};
% Signal strength in dBm
signalStrength = [-82 -76 -69 -61 -96 -101 -73 -66];

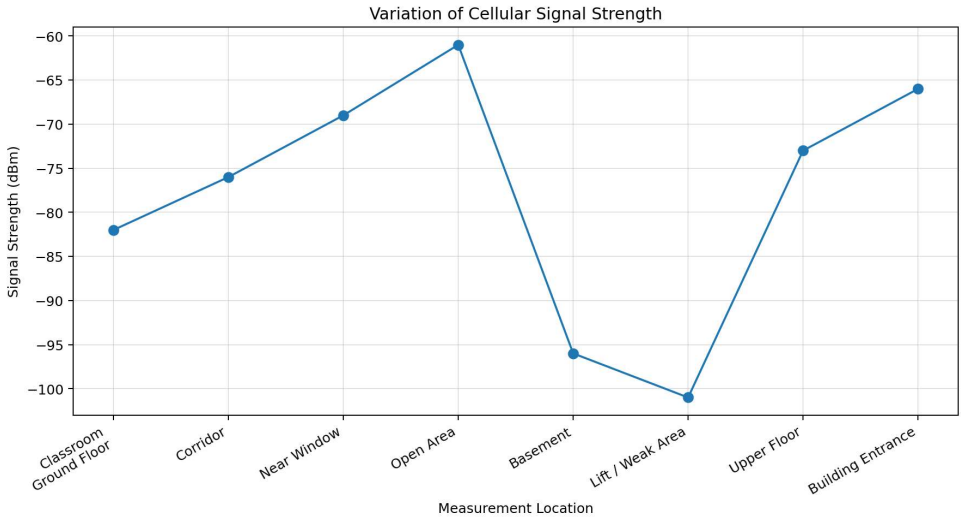
% Display data
T = table(location', networkType', signalStrength', ...
 'VariableNames', {'Location','Network_Type','Signal_Strength_dBm'});
disp(T);
% Average signal strength
avgSignal = mean(signalStrength);
fprintf('Average signal strength = %.2f dBm\n', avgSignal);

% Bar graph
figure;
bar(signalStrength);
grid on;
xticks(1:length(location));
xticklabels(location);
xtickangle(35);
xlabel('Measurement Location');
```

```
ylabel('Signal Strength (dBm)');
title('Cellular Signal Strength at Different Locations');
% Line graph
figure;
plot(1:length(signalStrength), signalStrength, '-o', ...
'LineWidth', 1.5, 'MarkerSize', 7);
grid on;
xticks(1:length(location));
xticklabels(location);
xtickangle(35);
xlabel('Measurement Location');
ylabel('Signal Strength (dBm)');
title('Variation of Cellular Signal Strength');
```

Output :





Result : Cellular network parameters were measured using a smartphone at different indoor and outdoor locations. The network type and received signal strength varied with location. The sample measurements demonstrate observations involving 2G, 3G, 4G/LTE and 5G technologies and show that the radio environment affects received signal strength.

Conclusion:

The experiment demonstrates how a smartphone can be used to study cellular network conditions. Signal strength varies because of distance from the serving cell, building penetration, obstructions, frequency band, interference and network configuration. The network type may also change between 5G, 4G/LTE, 3G and 2G depending on coverage, operator configuration and device conditions. MATLAB provides a convenient method for organizing, averaging and visualizing the collected measurements.

Viva Voce Questions :

1. Why does cellular signal strength vary inside a building?
2. Why can a smartphone change from 5G to 4G/LTE or another technology?

MARKS & SIGNATURE :

R1 (3 Marks)	R2 (5 Marks)	R3 (4Marks)	R4 (3 Marks)	Total (15 Marks)	Signature